

Matrix functions and stability

Temporal regularity of marginal matrix-product growth

Difficulty: challenging **Importance:** interesting to specialist

Status: open; two-part question retained as one entry

Context and notation

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d \times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

For a compact nonempty $\mathcal{M} \subset \mathbb{R}^{d \times d}$ with $\hat{\rho}(\mathcal{M}) = 1$, define its maximal product norm at length k by

$$g_{\mathcal{M}}(k) = \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2.$$

This problem concerns growth within one family over time. [MF-07](#) instead requests a dimension-dependent bound uniform across families.

Problem statement

Does every family in this notation satisfy both

$$\exists c > 0 \forall k, \ell \geq 1 : \quad g_{\mathcal{M}}(k + \ell) \geq c g_{\mathcal{M}}(k)$$

and

$$\forall \ell \geq 1 \exists C_{\ell} > 0 \forall k \geq 1 : \quad g_{\mathcal{M}}(\ell k) \leq C_{\ell} g_{\mathcal{M}}(k)?$$

Constants may depend on $\mathcal{M}$.

Reference and status evidence

Varney and Morris, [On marginal growth rates of matrix products](#), Linear Algebra Appl. 709 (2025), 132–163, [DOI](#), §4, Definition 4.1 and §7, Question 1. The source names these properties weak increase and weak upper regular variation. Searches for both property names, the authors, and subsequent matrix-product growth papers located no resolution.